

Predictive model assessment and selection in composite-based modeling using PLS-SEM: extensions and guidelines for using CVPAT

Pratyush N. Sharma

*Department of Information Systems, Statistics and Management Science,
The University of Alabama, Tuscaloosa, Alabama, USA*

Benjamin D. Liengaard

*Department of Economics and Business Economics, Aarhus University,
Aarhus, Denmark*

Joseph F. Hair

Department of Marketing, University of South Alabama, Mobile, Alabama, USA

Marko Sarstedt

*Munich School of Management, Ludwig-Maximilians-University Munich, Munich,
Germany and Faculty of Economics and Business Administration,
Babeş-Bolyai University, Cluj-Napoca, Romania, and*

Christian M. Ringle

*Department of Management Sciences and Technology,
Hamburg University of Technology, Hamburg, Germany*

Abstract

Purpose – Researchers often stress the predictive goals of their partial least squares structural equation modeling (PLS-SEM) analyses. However, the method has long lacked a statistical test to compare different models in terms of their predictive accuracy and to establish whether a proposed model offers a significantly better out-of-sample predictive accuracy than a naive benchmark. This paper aims to address this methodological research gap in predictive model assessment and selection in composite-based modeling.

Design/methodology/approach – Recent research has proposed the cross-validated predictive ability test (CVPAT) to compare theoretically established models. This paper proposes several extensions that broaden the scope of CVPAT and explains the key choices researchers must make when using them. A popular marketing model is used to illustrate the CVPAT extensions' use and to make recommendations for the interpretation and benchmarking of the results.

Findings – This research asserts that prediction-oriented model assessments and comparisons are essential for theory development and validation. It recommends that researchers routinely consider the application of CVPAT and its extensions when analyzing their theoretical models.



Research limitations/implications – The findings offer several avenues for future research to extend and strengthen prediction-oriented model assessment and comparison in PLS-SEM.

Practical implications – Guidelines are provided for applying CVPAT extensions and reporting the results to help researchers substantiate their models' predictive capabilities.

Originality/value – This research contributes to strengthening the predictive model validation practice in PLS-SEM, which is essential to derive managerial implications that are typically predictive in nature.

Keywords Evaluation, PLS, Prediction, SEM, Model selection, Partial least squares, Assessment, Structural equation modeling, Cross-validation

Paper type Research paper

Introduction

Model comparisons are an essential research tool for documenting the evolution of theoretical models and assessing whether the understanding of a phenomenon has improved (Sharma *et al.*, 2019). Comparing alternative models places theories under sharp scrutiny and enables comparative testing to confirm or falsify theories (Gray and Cooper, 2009; Popper, 1959). Empirically, two different approaches have been proposed for conducting model comparisons. One approach is explanation, which focuses on comparing how well the observed data fit the competing models via in-sample measures such as goodness-of-fit, explained variance (e.g. R^2) and information criteria, such as the Akaike and Bayesian information criteria (Sharma *et al.*, 2021; Hair *et al.*, 2017). The second approach is prediction which flips the relationship between the models and the data, so that the main focus is on comparing the accuracy (or error) of predicting data the models have not previously analyzed and relies on out-of-sample prediction techniques such as cross-validation (Hofman *et al.*, 2017; Yarkoni and Westfall, 2017). In other words, whereas the goal of explanation-oriented model comparison is to assess which model best explains the sample relationships at hand, the goal of the predictive approach is to assess which model more accurately predicts (holdout) data not previously used to optimize model fit (Shmueli, 2010). Moreover, it is important to examine both approaches since majority of management models carry both explanatory and predictive aspects (Gregor, 2006).

Statistical methods for estimating hypothesized models differ in their ability to accommodate in-sample explanation and out-of-sample prediction-oriented model comparisons. For example, in the context of estimating models with latent and observed variables, covariance-based structural equation modeling (SEM) primarily follows the explanation-oriented paradigm by focusing on goodness-of-fit measures (Hair *et al.*, 2017). In contrast, partial least squares structural equation modeling (PLS-SEM) was developed as a “causal-predictive” approach (Jöreskog and Wold, 1982, p. 270) to enable simultaneous explanation and prediction-oriented model assessments (Chin *et al.*, 2020; Hair *et al.*, 2019b, Sharma *et al.*, 2022). Despite its dual focus, PLS-SEM has primarily been used in explanation-oriented studies that have paid lip service to prediction due to a lack of suitable prediction-oriented tools (Shmueli *et al.*, 2016). This observation also hold true for the marketing discipline (Sarstedt *et al.*, 2022).

To address this issue, Liengaard *et al.* (2021) recently proposed the *cross-validated predictive ability test* (CVPAT), which was developed to compare the predictive ability of two theoretically motivated models by means of an overall model-level statistical inference test. The test focuses on predicting *all* endogenous constructs simultaneously, thereby considering the entire model's predictive accuracy. As such, CVPAT currently lacks two critical capabilities:

- (1) the ability to compare a single proposed model with a naïve benchmark to ensure it meets minimum predictive accuracy; and
- (2) the ability to compare two models by focusing on the prediction accuracy of specific constructs, and not necessarily of all the constructs simultaneously.

Predictive benchmarking is important because it enables researchers to assess whether their proposed model is able to improve on a naïve baseline, a key requirement for predictive validity (Shmueli and Koppius, 2011). The ability to compare two models at the construct level (based on the prediction of a subset of constructs) is also important to develop managerial strategies designed to optimize key outcomes, such as customer satisfaction and loyalty.

This paper fills the characterized research gap by extending the *CVPAT* framework to enable predictive benchmarking as well as comparing the predictive accuracy of models at the construct level, thereby complementing the overall model-level comparison introduced by Lienggaard *et al.* (2021). To distinguish between the proposed extensions, Lienggaard *et al.* (2021) original *CVPAT* is referred as $CVPAT_{overall}^{compare}$ throughout this article, where the superscript *compare* denotes that the predictive comparison is between two models and the subscript *overall* indicates that the comparison occurs at the overall model level (i.e. by assessing the predictive accuracy of all indicators related to endogenous constructs in the model). The first extension consists of a set of methods that enable assessing the predictive accuracy of a single proposed model against two criteria:

- (1) a naïve indicator-averages prediction benchmark (denoted by the superscript *benchmark_IA*) both on the *overall* model level ($CVPAT_{overall}^{benchmark_IA}$) and for selected *construct(s)* ($CVPAT_{construct}^{benchmark_IA}$); as well as
- (2) a conservative linear model prediction benchmark (denoted by the superscript *benchmark_LM*) both on the *overall* model level and for selected *construct(s)* ($CVPAT_{overall}^{benchmark_LM}$ and $CVPAT_{construct}^{benchmark_LM}$).

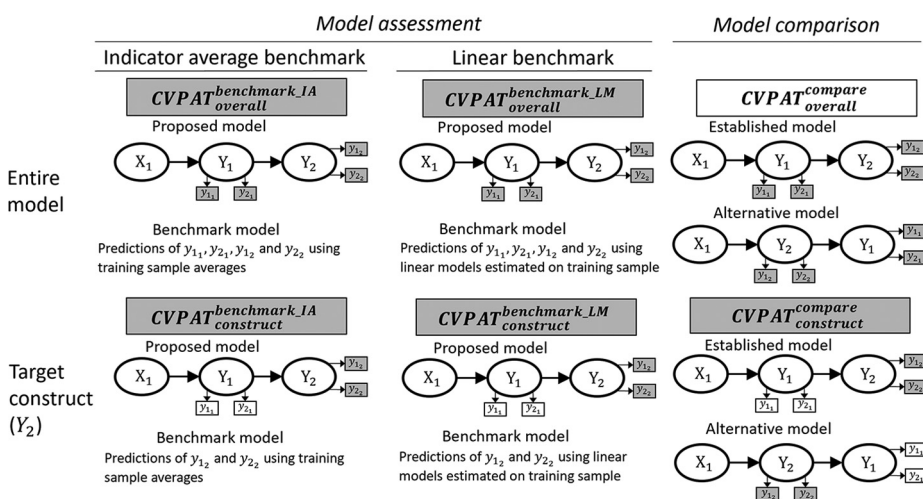
The capabilities of the $CVPAT_{overall}^{compare}$ are further extended by introducing the $CVPAT_{construct}^{compare}$, which enables researchers to compare two models at the *construct*-level based on the prediction errors of a subset of indicators related to one or more target constructs. Using a simple PLS path model with three constructs as example, Figure 1 shows the extended *CVPAT* framework and the relative positioning of the extensions proposed in this paper, *vis-à-vis* the original $CVPAT_{overall}^{compare}$ proposed by Lienggaard *et al.* (2021).

The following sections provide a background on the $CVPAT_{overall}^{compare}$, introduce the extensions of this approach and provide guidelines for applying the extended *CVPAT* framework and reporting the results. The application of the framework and guidelines for interpreting results are illustrated using the well-known corporate reputation model (Hair *et al.*, 2022). The final section presents conclusions and avenues for future research.

Background and extensions

The cross-validated predictive ability test

The original $CVPAT_{overall}^{compare}$ provides a statistical test to compare the predictive accuracy of an established model with that of an alternative model. To achieve this aim, $CVPAT_{overall}^{compare}$ relies on the cross-validation technique, which randomly divides the dataset into a number of subsamples (i.e. folds) that iteratively serve as training and holdout sets (Lienggaard *et al.*, 2021). In the first iteration, the first fold is reserved as an independent holdout set and the model is estimated on the remaining observations acting as the training set. The output variables of the first fold are then predicted using the input variables of the first fold together with the parameter estimates obtained from the training set data. The procedure is repeated for each fold to generate an out-of-sample prediction, $\tilde{y}_{i,k_q}$, for each *individual*



Notes: $CVPAT_{overall}^{compare}$ denotes Liengard *et al.*'s (2021) original CVPAT approaches shaded in gray indicate extensions proposed in this paper. Indicators shaded in gray are those used in model assessment/comparison

Figure 1.
The extended
CVPAT framework

observation (e.g. a survey respondent) $i = 1, \dots, N$, and for each indicator, $k_q = 1, \dots, k_q$, related to the *endogenous* constructs $q = 1, \dots, Q$. The actual values, y_{i,k_q} , are subtracted from the predicted values to obtain prediction errors:

$$\tilde{e}_{i,k_q} = y_{i,k_q} - \hat{y}_{i,k_q}. \quad (1)$$

If $\mathcal{M} = \{1_1, \dots, K_1, \dots, 1_q, \dots, K_q, \dots, 1_Q, \dots, K_Q\}$ is a set containing all indices of the indicators related to endogenous constructs, then $CVPAT_{overall}^{compare}$ summarizes the loss for individual i as:

$$L_i = \frac{1}{|\mathcal{M}|} \sum_{p \in \mathcal{M}} (\tilde{e}_{i,p})^2, \quad (2)$$

where $|\mathcal{M}|$ is the total number of indicators related to the endogenous constructs. To compare two models with each other, Liengard *et al.* (2021) use the average loss difference

$$\bar{D} = \frac{1}{N} \sum_{i=1}^N (L_i^{EM} - L_i^{AM}), \quad (3)$$

where L_i^{EM} and L_i^{AM} are the individual's losses associated with the established and alternative models, respectively. The null and alternative hypotheses are:

$$\begin{aligned} H_0 : E(\bar{D}) &= Loss^{EM} - Loss^{AM} = 0 \text{ and} \\ H_1 : E(\bar{D}) &= Loss^{EM} - Loss^{AM} > 0, \end{aligned} \quad (4)$$

where E denotes the expected value and $Loss^{EM} = E(L_i^{EM})$ and $Loss^{AM} = E(L_i^{AM})$. Hypothesis testing is conducted using the test statistic $T = \frac{\bar{D}}{\sqrt{S^2/N}}$, where $S^2 = \frac{1}{N-1} \sum_{i=1}^N (D_i - \bar{D})^2$ and $D_i = L_i^{EM} - L_i^{AM}$. Through an extensive simulation study, Liengard *et al.* (2021) show this test statistic provides adequate Type I error rates and relevant power levels when evaluated against a t -distribution with $N - 1$ degrees of freedom. Alternatively, bootstrapping can be used for significance testing.

The concept of loss plays a central role in $CVPAT_{overall}^{compare}$ and its extensions and is a measure of the difference between predicted values and actual values. A loss value is calculated for each individual (e.g. a loss for a respondent) as the average squared prediction error over all indicators associated with the endogenous constructs. The overall loss of a model is the average of all the individual losses. A low overall loss indicates the model predicts the out-of-sample observations well. In model comparisons, a model with lower loss is considered superior in terms of out-of-sample prediction performance.

Benchmarking extensions for predictive model assessment

Consistent with Shmueli *et al.* (2019), the extensions introduced in this paper allow comparing a single proposed model with both the IA and LM prediction benchmarks to assess its predictive accuracy. When the focus is on the simultaneous prediction of all endogenous constructs in a PLS path model, $CVPAT_{overall}^{benchmark\ IA}$ allows for comparisons of a proposed model with the IA benchmark, whereas $CVPAT_{overall}^{benchmark\ LM}$ allows for the comparison with the LM prediction benchmark. The IA benchmark uses the indicator averages from the training sample as predicted values, while the LM benchmark uses the holdout observations together with the linear regression coefficients, estimated from the training sample, to obtain the predicted values.

To enable these comparisons, the prediction error in the naïve IA prediction benchmark model is calculated as follows:

$$\tilde{e}_{i,k_q}^{IA} = y_{i,k_q} - \tilde{y}_{i,k_q}^{IA}, \quad (5)$$

where $\tilde{y}_{i,k_q}^{IA}$ is the training sample average of indicator y_{k_q} (i.e. the average of indicator y_{k_q} across all observations except those in the same fold as observation i). Similarly, when using the LM prediction benchmark, the prediction errors are obtained as follows:

$$\tilde{e}_{i,k_q}^{LM} = y_{i,k_q} - \tilde{y}_{i,k_q}^{LM}, \quad (6)$$

where $\tilde{y}_{i,k_q}^{LM}$ is the out-of-sample prediction of y_{i,k_q} using a linear model [1]. Using $\tilde{e}_{i,k_q}^{IA}$ or $\tilde{e}_{i,k_q}^{LM}$ as prediction errors in equation (2), individual losses can be obtained for the IA benchmark (L_i^{IA}) and the LM benchmark (L_i^{LM}) respectively. The average loss difference for the indicator average benchmark is given by:

$$\bar{D}^{IA} = \frac{1}{N} \sum_{i=1}^N (L_i^{proposed} - L_i^{IA}), \quad (7)$$

where $L_i^{proposed}$ is the individual loss associated with the proposed PLS path model. To denote the corresponding average loss difference in the LM prediction benchmark the

superscript *IA* in [equation \(7\)](#) is replaced by *LM*. For hypothesis testing, the following null and alternative hypotheses are used:

$$\begin{aligned} H_0 : E(\bar{D}^{IA}) &= Loss^{proposed} - Loss^{IA} = 0 \text{ and} \\ H_1 : E(\bar{D}^{IA}) &= Loss^{proposed} - Loss^{IA} < 0, \end{aligned} \quad (8)$$

where $Loss^{proposed} = E(L_i^{proposed})$ and $Loss^{IA} = E(L_i^{IA})$.

Again, the equivalent hypothesis of the LM prediction benchmark is obtained by substituting the superscript *IA* with *LM* in [equation \(8\)](#) as follows:

$$\begin{aligned} H_0 : E(\bar{D}^{LM}) &= Loss^{proposed} - Loss^{LM} = 0 \text{ and} \\ H_1 : E(\bar{D}^{LM}) &= Loss^{proposed} - Loss^{LM} < 0, \end{aligned} \quad (9)$$

where $Loss^{LM} = E(L_i^{LM})$

These extensions allow researchers to assess whether their proposed model has superior predictive accuracy in relation to the naive IA and LM prediction benchmarks. They can use this analysis for all endogenous constructs or for certain selected endogenous constructs in the model ([Figure 1](#)) – in a manner similar to the extension we discuss in the next section.

Predictive model comparison extension for selected endogenous constructs

[Lienggaard et al.'s \(2021\)](#) original $CVPAT_{overall}^{compare}$ enables the simultaneous comparison of two theoretically motivated models based on the predictive accuracy of all endogenous constructs in the models. In many instances, however, the research focus is on the predictive accuracy of specific target constructs ([Shmueli et al., 2019](#)), and not necessarily on all the endogenous constructs in the model. For example, in the American customer satisfaction index model (ACSI; [Fornell et al., 1996](#); [Anderson and Fornell, 2000](#)) and the European customer satisfaction model (ECSI; [Eklöf and Westlund, 2002](#); [Ball et al., 2004](#)), the customer complaint behavior construct serves as a mediator of the relationship between customer satisfaction and customer loyalty, and is thus endogenous. However, the aim is not to predict customer complaint behavior. Instead, the focus is on the prediction of the key target construct customer satisfaction and – to a lesser degree – customer loyalty.

For this purpose, the proposed $CVPAT_{construct}^{compare}$ enables the comparison of models based on the predictive accuracy of a subset of endogenous constructs. Let *C* denote a set of selected target constructs (or a single endogenous construct), and $\mathcal{M}(C)$ be the set of indices for the indicators related to these constructs, then [equation \(2\)](#) can be modified to summarize the loss of individual *i* based on the set *C* as follows:

$$L_i(C) = \frac{1}{|\mathcal{M}(C)|} \sum_{p \in \mathcal{M}(C)} (\tilde{e}_{i,p})^2. \quad (10)$$

After all individual losses have been calculated according to [equation \(10\)](#), L_i can be substituted with $L_i(C)$ in [equation \(3\)](#) to obtain the average loss difference on the set *C* as follows:

$$\bar{D}(C) = \frac{1}{N} \sum_{i=1}^N (L_i^{EM}(C) - L_i^{AM}(C)). \quad (11)$$

Equation (11) allows focusing on selected target constructs in the PLS path model and to compute a test statistic for predictive model comparison in the same way as done by Liengaard *et al.* (2021). More importantly, $CVPAT_{construct}^{compare}$ can also be used in conjunction with the IA or LM prediction benchmarks to assess a single model's predictive accuracy with regard to specific endogenous constructs (i.e. by using $CVPAT_{construct}^{benchmark_{IA}}$ or $CVPAT_{construct}^{benchmark_{LM}}$; Figure 1). To conduct these model assessments, the procedure described in the previous section can be followed but only by considering the prediction errors of the indicators associated with the selected target endogenous constructs (i.e. indicators with indices in the set $\mathcal{M}(C)$). The next section offers researchers guidance regarding the systematic use of the extended *CVPAT* framework in their research studies [2].

Guidelines for the application of the extended cross-validated predictive ability test framework

The application of the extended *CVPAT* framework for prediction-oriented model assessment and comparison in PLS-SEM requires a number of decisions by researchers. Figure 2 shows the proposed guidelines.

In Stage 1, the researcher decides whether the analysis focuses on assessing the predictive accuracy of all endogenous constructs in the model simultaneously or of selected key constructs only. Stage 2 requires prediction-oriented model assessment based on the IA and LM prediction benchmarks to determine the model(s) level of predictive accuracy. In the final (optional) step, depending on the goal of the study, the researcher may undertake predictive model comparison of two theoretically motivated models. These stages are described in more detail below.

Selection of target constructs (Stage 1)

The researcher has to decide whether the analysis focuses on assessing the predictive accuracy of *all* endogenous constructs in the model simultaneously (i.e. the overall model), or of only *selected* target constructs. For example, the goal of the unified theory of acceptance and use of technology (UTAUT; Venkatesh *et al.*, 2003) and its successor UTAUT2 (Venkatesh *et al.*, 2012) is the simultaneous prediction of two endogenous constructs – behavioral intention and use behavior. In contrast, the ACSI and ECSI models focus on the target constructs customer satisfaction and customer loyalty, while the other endogenous construct, customer complaints, serves as a mediator and is not the focus of the prediction. Based on these considerations, the researcher decides whether to use the $CVPAT_{overall}^{compare}$ or the $CVPAT_{construct}^{compare}$ in the subsequent stages of executing the model assessment and

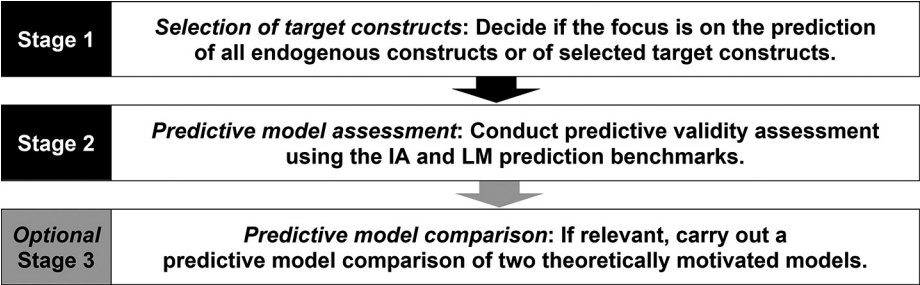


Figure 2.
Guidelines for using
the extended *CVPAT*
framework

comparison. It is important to note that the focal decision regarding which constructs to optimize for prediction is made *a priori* based on theory and research goal. In addition, model selection based on multiple post hoc analyses must be done cautiously since it could lead to type I errors (Liengard *et al.*, 2021).

Predictive model assessment (Stage 2)

In the second step, the researcher assesses the predictive accuracy of the model(s) using indicator averages (i.e. $CVPAT_{overall}^{benchmark_IA}$ or $CVPAT_{construct}^{benchmark_IA}$) and linear model prediction benchmarks (i.e. $CVPAT_{overall}^{benchmark_LM}$ or $CVPAT_{construct}^{benchmark_LM}$). Figure 3 displays the steps involved in Stage 2 and the possible conclusions that can be drawn.

As shown in Figure 3, the IA prediction benchmark is less conservative and is used first. Table 1 depicts the statistical inference procedure for predictive model assessment. To have predictive validity, the model(s) should have a significantly lower average loss (i.e. higher predictive accuracy) than the naïve IA benchmark. If this is not the case, the model(s) does not have predictive validity and must be discarded. If the model(s) outperforms the IA prediction benchmark, it can be concluded that it meets the minimum standard for predictive validity. The researcher can then compare it with the conservative LM prediction benchmark (Shmueli *et al.* (2019). If the model(s) exhibits significantly lower average loss (i.e. higher predictive accuracy) than the LM prediction benchmark, the conclusion is that it has strong predictive validity (Figure 3).

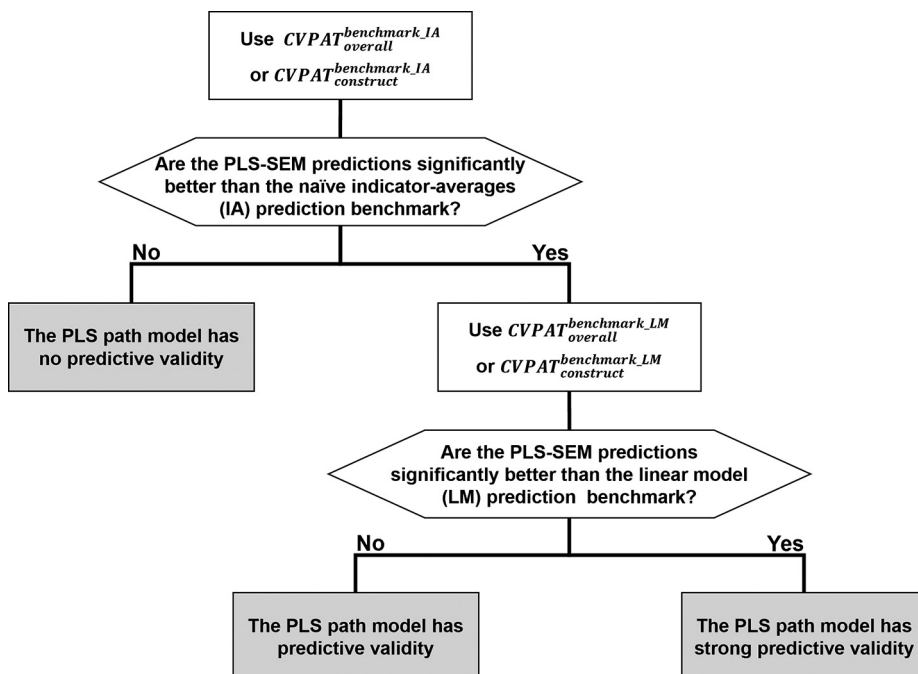


Figure 3.
CVPAT
benchmarking for
predictive model
assessment (Stage 2)

Table 1.
Statistical test
(Stage 2)

Benchmarks	
Indicator-averages (IA): $CVPAT_{overall}^{benchmark_IA}$ or $CVPAT_{construct}^{benchmark_IA}$	or $CVPAT_{construct}^{benchmark_LM}$
Linear model (LM): $CVPAT_{overall}^{benchmark_LM}$ or $CVPAT_{construct}^{benchmark_LM}$	
Hypotheses	Test outcome and conclusion
$H_0: Loss^{proposed} = Loss^{benchmark}$	Cannot reject $H_0 \rightarrow$ Predictive validity not confirmed (PLS path model does not outperform the benchmark)
$H_1: Loss^{proposed} < Loss^{benchmark}$	Reject $H_0 \rightarrow$ Predictive validity is supported (PLS path model outperforms the benchmark)
Note: $Loss^{benchmark}$ refers to either $Loss^{IA}$ or $Loss^{LM}$	

Predictive model comparison (Stage 3)

The optional Stage 3 of the proposed procedure addresses predictive model comparisons between two theoretically motivated models, of which one serves as the established model and the other as the proposed or alternative model (Figure 2). Each of the two models assessed during this stage must exhibit predictive validity by surpassing the IA benchmark. If the models are simultaneously compared for the predictive accuracy of all endogenous constructs, the researcher should use $CVPAT_{overall}^{compare}$. If the focus is on the prediction of selected target constructs, then $CVPAT_{construct}^{compare}$ should be used.

Table 2 shows the statistical inference process. The alternative model is only favored if it exhibits significantly lower average loss than the established model. If not, the established model is retained as the best predictive model (for further details and explications, see Liengaard et al., 2021).

Empirical illustration: the corporate reputation model. The corporate reputation model originally presented by Eberl (2010) is used to illustrate the application of the extended CVPAT framework. The aim of this model is to explain the effects of competence (COMP) and likeability (LIKE), the two core dimensions of corporate reputation (Schwaiger, 2004; Sarstedt et al., 2013), on customer satisfaction (CUSA) and customer loyalty (CUSL). In addition, the model includes the quality of customer orientation (QUAL), economic and managerial performance (PERF), corporate social responsibility (CSOR) and the attractiveness (ATTR) of a company as four antecedents of the corporate reputation dimensions, COMP and LIKE (see the established model, EM, in Figure 4). In addition to the numerous marketing applications (Zhang and Schwaiger, 2012), the corporate reputation

Table 2.
Statistical test
(Stage 3)

Focus of model comparison	
Predictive accuracy of all endogenous constructs: Use $CVPAT_{overall}^{compare}$	
Predictive accuracy of key target construct(s): Use $CVPAT_{construct}^{compare}$	
Hypotheses	Test outcome and conclusion
$H_0: Loss^{AM} = Loss^{EM}$	Cannot reject $H_0 \rightarrow$ Retain EM
$H_1: Loss^{AM} < Loss^{EM}$	Reject $H_0 \rightarrow$ Select AM
Notes: EM = established model; AM =alternative model	

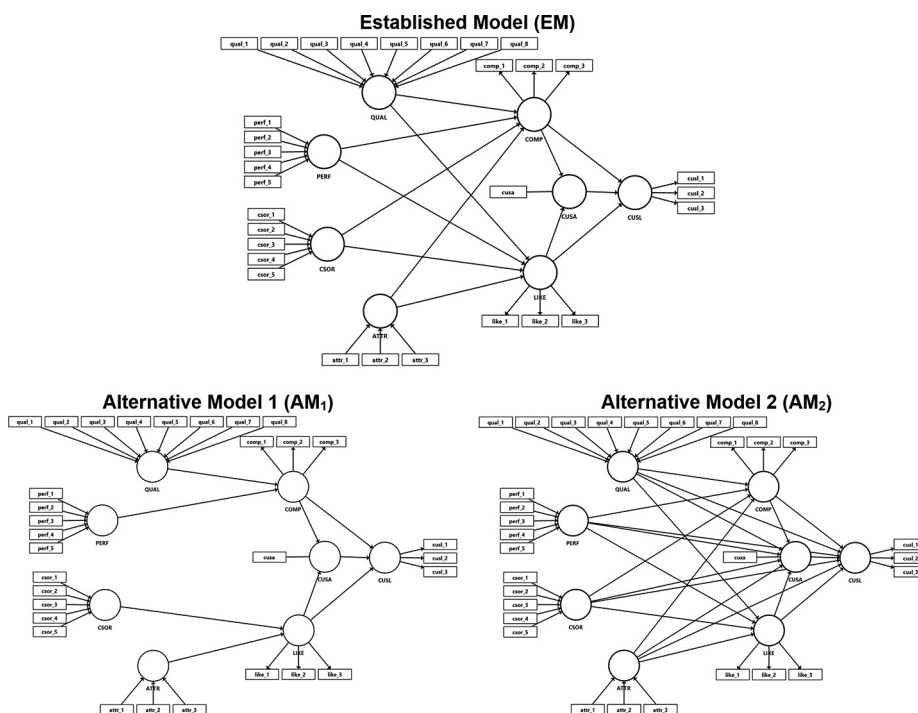


Figure 4.
The corporate
reputation model and
alternatives

model has been used extensively to illustrate the capabilities and methodological advantages of the PLS-SEM method (Hair *et al.*, 2018; Hult *et al.*, 2018; Matthews *et al.*, 2016). This illustration uses the data set that Hair *et al.* (2022) and Sarstedt *et al.* (2017) used to present PLS-SEM estimation results and their evaluation. The model estimations draw on 344 responses collected from two major German mobile communications network providers and two smaller competitors. The assessment of the measurement and structural models show that they meet the relevant criteria (for further details of the PLS-SEM results and their assessment, Hair *et al.*, 2022; Sarstedt *et al.*, 2017).

In Stage 1, the target constructs for the analysis are selected. This illustration focuses on *CUSA* and *CUSL*, because these constructs are the basis for drawing managerial implications in the customer satisfaction and loyalty literature (Fornell *et al.*, 1996; Anderson and Fornell, 2000) [3].

In Stage 2, the predictive validity of the corporate reputation model is assessed. To begin, $CVPAT_{construct}^{benchmark_IA}$ is used to focus on the endogenous constructs *CUSA* and *CUSL*. The results indicate that the corporate reputation model has significantly lower average loss (i.e. lower prediction error) than the IA prediction benchmark (Table 3).

In the next step of the predictive model assessments using *CVPAT*, the conservative LM prediction benchmark is used (Figure 3). While the model exhibits lower average loss than the LM prediction benchmark, the difference is not statistically significant at the 0.05 level (Table 3). Hence, it can be concluded that although the corporate reputation model meets the minimum standards of predictive validity (i.e. it beats the IA

benchmark), it does not exhibit strong predictive validity (i.e. it does not beat the LM benchmark).

Stage 3, which is optional, is used if the goal of the study is to choose between two theoretically established model alternatives with established predictive validity. For illustrative purposes, the corporate reputation model (i.e. the established model; EM) is compared with two alternative models. Following [Sharma et al. \(2021\)](#), a parsimonious alternative model (AM₁) and a complex alternative model (AM₂) are considered, as shown in [Figure 4](#) (see also [Danks et al., 2020](#)). AM₁ proposes that performance (*PERF*) and quality (*QUAL*) are the only antecedents of the cognitive corporate reputation dimension competence (*COMP*), while attractiveness (*ATTR*) and corporate social responsibility (*CSOR*) are the only antecedents of the affective corporate reputation dimension likeability (*LIKE*). In contrast, the complex model AM₂ considers all possible relationships. As a result, the *R*² values of *CUSA* and *CUSL* constructs are expected to be higher in AM₂ than in EM or AM₁. The specific prediction question of interest is whether this alternative model also offers a higher predictive accuracy than the EM.

Before starting the analysis, it should be noted that both of the alternative models met all relevant measurement and structural evaluation criteria ([Hair et al., 2019a](#)). However, as in the EM, the complex model AM₂ also has some statistically nonsignificant relationships in the structural model. As in the previously published analyses of the EM ([Hair et al., 2022](#); [Sarstedt et al., 2017](#)), these relationships are retained in the model. The lack of significance of these relationships can be attributed to the specific context of model estimation (i.e. German mobile telephone providers). As in Stage 2 ([Figure 3](#)), $CVPAT_{construct}^{benchmark_IA}$ and $CVPAT_{construct}^{benchmark_LM}$ are used to confirm that AM₁ meets the minimum standards of predictive validity (i.e. it beats the IA benchmark), while AM₂ exhibits strong predictive validity (i.e. it beats both the IA and LM benchmarks).

Having established the predictive validity of all three models (i.e. EM, AM₁ and AM₂), the Stage 3 of the analysis compares the EM with AM₁ and AM₂, respectively. As shown in [Table 4](#), the $CVPAT_{construct}^{compare}$ results indicate the average losses of the selected target constructs *CUSA* and *CUSL* are lower in the EM than in both AM₁ and AM₂. This finding supports the original corporate reputation model (EM)'s superior predictive accuracy over that of AM₁ and AM₂.

Table 3.
CVPAT Benchmark
results for predictive
model assessment

Focus on <i>CUSA</i> and <i>CUSL</i>	PLS-SEM loss	Benchmark loss	Difference	<i>p</i> -value
$CVPAT_{construct}^{benchmark_IA}$	0.629	1.015	−0.386	0.000
$CVPAT_{construct}^{benchmark_LM}$	0.629	0.638	−0.009	0.067

Note: loss = average loss

Table 4.
 $CVPAT_{construct}^{compare}$
Results of the
alternative models

Focus on <i>CUSA</i> and <i>CUSL</i>	EM loss	AM loss	Difference	<i>p</i> -value
$CVPAT_{construct}^{compare}$ of EM and AM ₁	0.629	0.630	−0.001	0.781
$CVPAT_{construct}^{compare}$ of EM and AM ₂	0.629	0.633	−0.004	0.681

Notes: EM = established model; AM₁ = alternative model 1; AM₂ = alternative model 2; loss = average loss

Conclusions and future research

PLS-SEM supports both explanatory and predictive analyses in path models (Cheah *et al.*, 2020; Hair *et al.*, 2019a). However, marketing researchers using the method typically restrict their analyses to explanatory model assessments by interpreting in-sample measures, such as R^2 (Liengard *et al.*, 2021; Sarstedt *et al.*, 2022). This is problematic in light of increasing calls that marketing researchers should complement their explanation-oriented analyses with prediction-orientation as is common in the natural sciences (Hofman *et al.*, 2017). In a similar vein, researchers have started acknowledging that predictive model assessments are crucial for enhancing practical and managerial implications (Sharma *et al.*, 2019; Shmueli and Koppius, 2011). To facilitate such analyses, recent research has introduced prediction-oriented tools for PLS-SEM, among which $PLS_{predict}$ (Shmueli *et al.*, 2016) and $CVPAT_{overall}^{compare}$ (Liengard *et al.*, 2021) are of particular note. Both approaches are restricted, however, in that they either focus on naïve benchmarking on the indicator-level (in the case of $PLS_{predict}$), or model comparisons of the overall model only (in the case of the $CVPAT_{overall}^{compare}$). Addressing this concern, this research extended the *CVPAT* framework to allow comparing models for their predictive accuracy at the construct level, as well as predictive benchmarking (Figure 1). By extending Liengard *et al.*'s (2021) $CVPAT_{overall}^{compare}$ to facilitate construct-level assessments, this study offers a suitable means for deriving managerial recommendations geared toward optimizing the prediction of a model's key target constructs. In addition, predictive benchmarking enables researchers to assess whether their model is able to beat the naïve IA prediction benchmark to ensure minimum predictive validity, or even the conservative LM prediction benchmark to ensure strong predictive validity. This assessment is important, because many well-established social science models are unable to beat even naïve prediction benchmarks, raising questions about their generalizability and practical use (Ward *et al.*, 2010). Model comparisons, on the other hand, facilitate identifying theoretically derived alternative models that extend or refine an established model. Such an assessment may increase the confidence in the established model, or it may be used as a starting point to further improve established models. Finally, a detailed presentation based on the well-known corporate reputation model illustrated the use of the proposed extensions and offers guidance regarding how to implement and report benchmarking and composite-based model comparison analyses.

Among other interesting future research opportunities (e.g. see Table 7 in Sarstedt *et al.*, 2022), the increasing relevance of predictive model assessments in PLS-SEM opens up several avenues for further methodological refinements (Chin *et al.*, 2020; Hair *et al.*, 2019b; Sharma *et al.* 2022). For instance, the question regarding whether and when researchers should consider the indicators of the earliest or direct predecessor constructs in prediction-oriented model assessments and comparisons remains unexplored (Danks, 2021). Identifying research settings that favor each approach would enable researchers to identify constructs and model structures that perform better or worse than others in predictive model comparisons. Future research should also assess the use of the extended *CVPAT* framework in the analysis of mediation effects (Sarstedt *et al.*, 2020; Nitzl *et al.*, 2016). When testing a model's predictive accuracy, the question arises whether only the exogenous constructs and their indicators should be used (Shmueli *et al.*, 2016). Alternatively, also constructs in a model that have both antecedents and successors (e.g. mediators) could be used with their indicators to predict subsequent endogenous variables. Progress in these directions would strengthen the knowledge on predictive model comparison options and capabilities in composite-based PLS-SEM, thereby enabling researchers to better use explanatory and predictive paradigms simultaneously (Sarstedt *et al.*, 2016). Another avenue for future research might be to weight prediction errors based on the number of indicators a construct has. Finally, future research should address how to assess the potential conflicts that

may arise when using the extended *CVPAT* framework on all or selected endogenous constructs simultaneously versus on each endogenous construct separately.

Notes

1. The prediction $\hat{y}_{i,k_q}^{LM}$ is constructed in two steps: using the individuals not in individual i 's fold, y_{k_q} is regressed on all indicators, which are the direct antecedents of construct q , and; using the estimated regression coefficients from Step 1 together with individual i 's values of the indicators' directly antecedent to construct q , the prediction $\hat{y}_{i,k_q}^{LM}$ is formed.
2. The R code for *CVPAT* and its extended framework proposed in this research is available online at: https://github.com/ECONshare/CVPAT_extensions
3. Alternatively, in this example, one could consider focusing on all endogenous constructs, as the model consists of the central explanators of the two corporate reputation dimensions *COMP* and *LIKE*, which in turn explain the key target constructs *CUSA* and *CUSL*. However, for illustrative purposes, we focus on the latter two constructs.

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About the authors

Pratyush N. Sharma is an Assistant Professor in the department of Information Systems, Statistics and Management Science in the University of Alabama's Culverhouse College of Business. His research interests include online collaboration communities, open source software development, technology use and adoption, and research methods used in information systems, particularly partial least squares path modeling. His research has been published in highly acclaimed journals such as the *Journal of the Association for Information Systems*, *Journal of Retailing*, *Decision Sciences*, *Journal of Information Systems*, *Journal of Business Research* and *Journal of International Marketing*.

Benjamin D. Liengaard is an Assistant Professor in the Department of Economics and Business Economics, Aarhus University. His main research interest is in partial least squares path modeling and quantitative analysis in the field of business analytics. His research has been published in *Decision Sciences*, the *European Journal of Marketing*, and *Psychology and Marketing*.

Joseph F. Hair, Jr. is Director of the PhD Program and Cleverdon Chair of Business, Mitchell College of Business, University of South Alabama. Since 2018, Joe was recognized by Clarivate Analytics for being in the top 1% globally of all Business and Economics professors based on his citations and scholarly accomplishments. He has authored over 75 book editions and has published numerous articles in scholarly journals such as the *Journal of Marketing Research*, *Journal of Academy of Marketing Science*, *European Journal of Marketing*, *Organizational Research Methods*, *Journal of Family Business Studies*, *Journal of Retailing* and others.

Marko Sarstedt is a Chaired Professor of Marketing at the Ludwig-Maximilians-University Munich, Germany, and an Adjunct Research Professor at Babeş-Bolyai University, Romania. His main research interest is the advancement of research methods to further the understanding of consumer behavior. His research has been published in *Nature Human Behaviour*, *Journal of Marketing Research*, *Journal of the Academy of Marketing Science*, *Multivariate Behavioral Research*, *Organizational Research Methods*, *MIS Quarterly*, *Decision Sciences* and *Psychometrika*, among others. Marko has been named member at Clarivate Analytics' Highly Cited Researchers List, which includes the "world's most impactful scientific researchers."

Christian M. Ringle is a Chaired Professor of Management at the Hamburg University of Technology, Germany. His research addresses human resource management, organization, marketing, strategic management and quantitative methods for business and market research and machine learning. His articles have been published in journals such as the *European Journal of Marketing*, *International Journal of Human Resource Management*, *International Journal of Research in Marketing*, *Journal of the Academy of Marketing Science*, *Organizational Research Methods* and *MIS Quarterly*. Since 2018, he has been continuously in the Clarivate Analytics' Highly Cited Researchers list. More information about Dr Ringle can be found at <https://www.tuhh.de/hrmo/team/prof-dr-c-m-ringle.html> Christian M. Ringle is the corresponding author and can be contacted at: c.ringle@tuhh.de

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